



IIT ROORKEE MOTORSPORTS

FULL CAR ACTUATION MODEL AND ANALYSIS OF CAMBER, TOE VARIATION ON VEHICLE PERFORMANCE

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OBJECTIVE

The objective of this project is to develop and analyze a full car actuation model for vehicle dynamics simulation, focusing on the effects of camber and toe variations on vehicle performance. This involves:

- Developing mathematical models and simulations in MATLAB/Simulink to evaluate longitudinal and lateral forces, ride comfort, handling stability, and tire wear under dynamic conditions.
- Building progressive models from quarter car to full car, incorporating anti-roll bars and analyzing bounce, pitch, roll, yaw, and wheel movements for optimized suspension parameters.

VEHICLE DYNAMICS MODELLING

Vehicle dynamics simulation is essential for optimizing suspension design and wheel geometry to improve ride comfort, handling stability, and tire longevity. This section details the modeling approaches, starting from tire forces and progressing to integrated vehicle models. Models are implemented in MATLAB, Simulink.

2.1 Tire Force Modeling

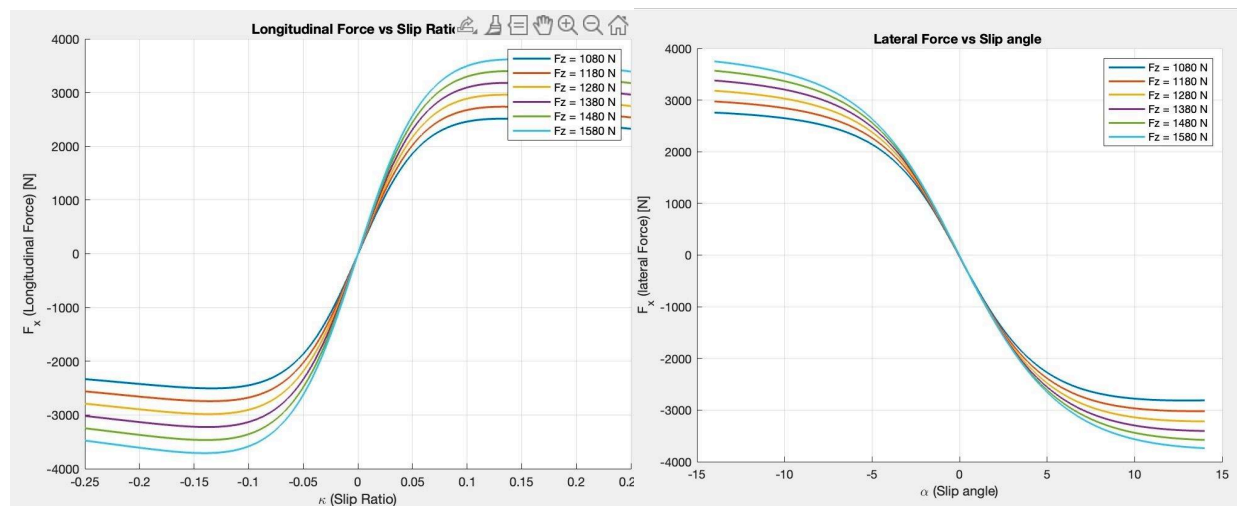
2.1.1 Variation of Longitudinal Force

At small slip values, the tire deforms elastically, and longitudinal force increases roughly linearly with slip ratio. The contact patch deforms but remains mostly in static friction with the road. As slip ratio increases, more of

the contact patch begins to slide, and the force increases at a slower rate, entering the nonlinear region where peak grip is approached.

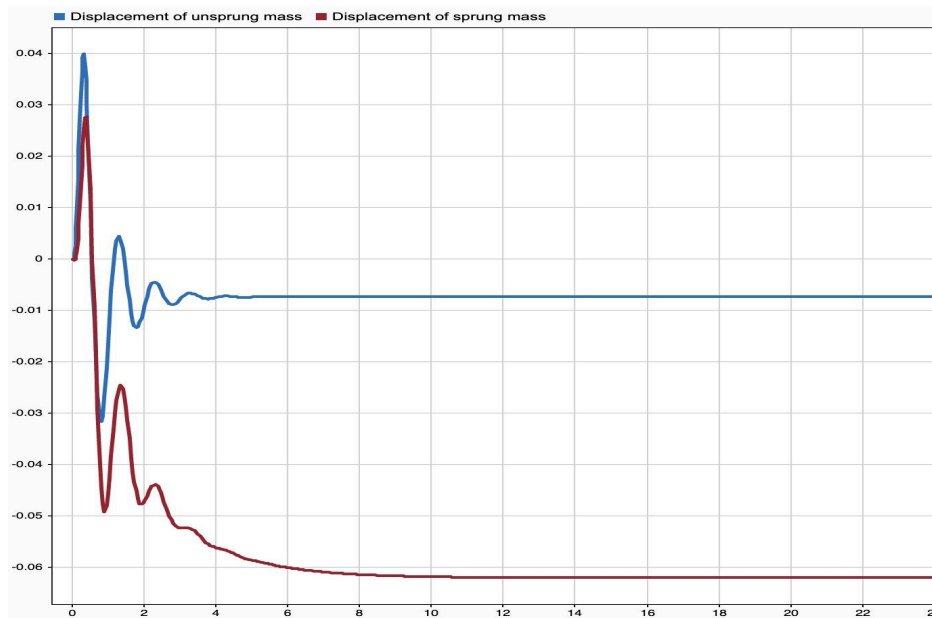
2.1.2 Variation of Lateral Force

When the tire moves exactly where it is pointed, there is no sideways stretch or slide, resulting in zero lateral force. As slip angle increases slightly, the tire's rubber bends sideways in the contact patch, creating a steady increase in sideways force. The initial slope is steeper with heavier loads (higher normal force F_z) due to increased contact area and pressure. Heavier loads boost maximum sideways force.



2.2 Quarter Car Model

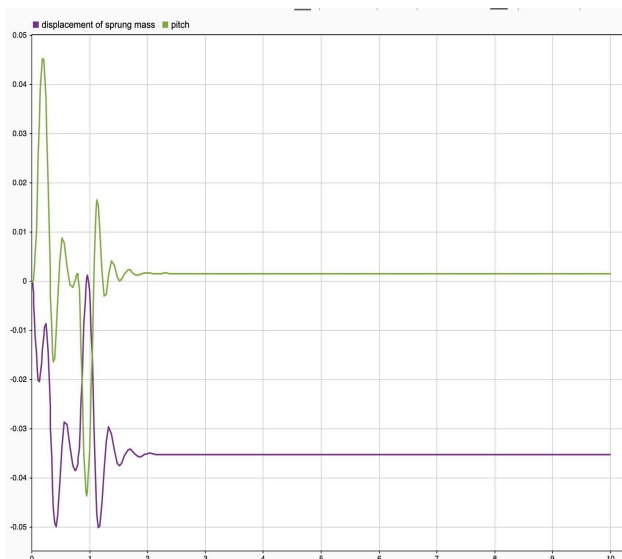
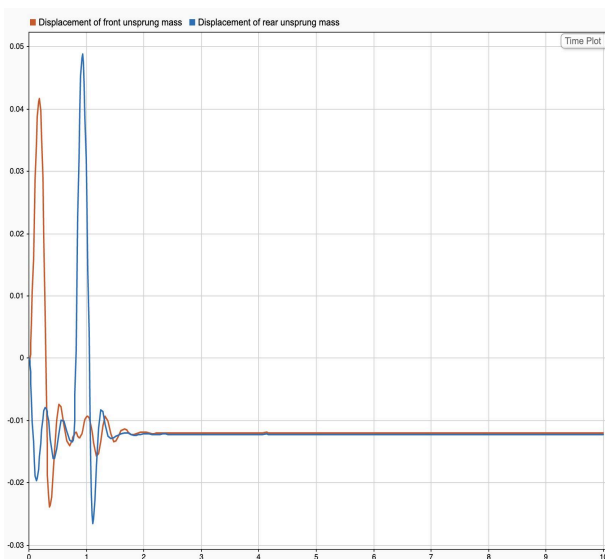
The quarter car model is a simplified 2-DOF suspension system focusing on vertical dynamics of one wheel. It simulates displacement of sprung (x_1 , vehicle body) and unsprung (x_2 , wheel) masses under a bump to minimize vertical accelerations transmitted to the body.



2.3 Halfcar model

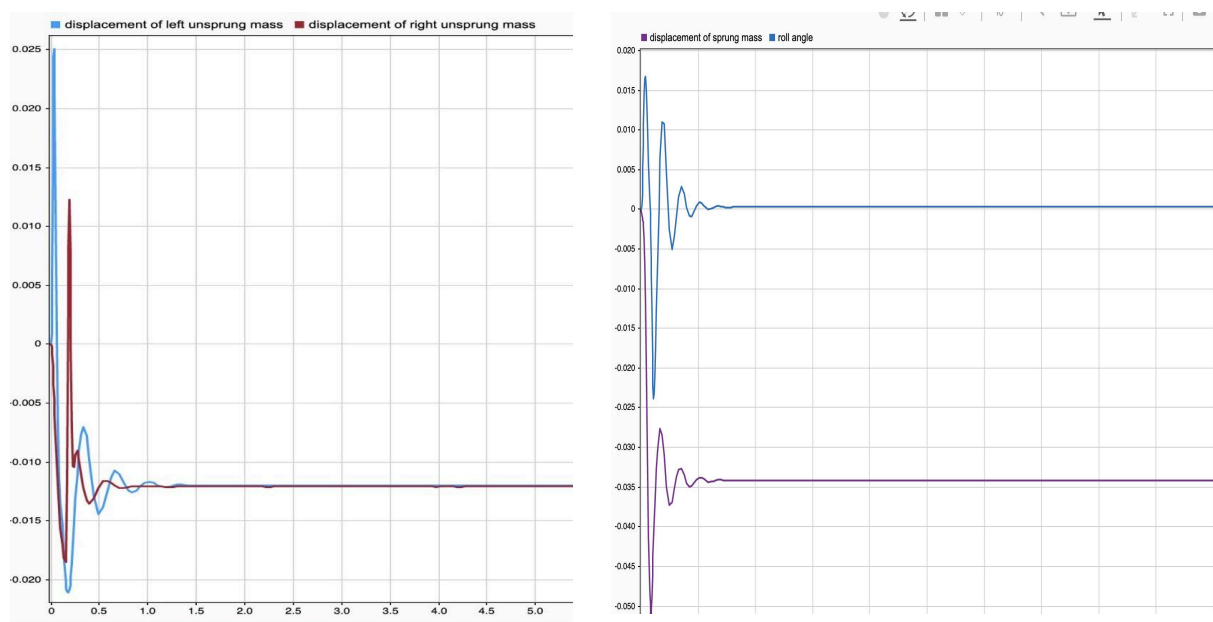
2.3.1 Pitch Analysis

The vehicle travels rightward, with the front wheel encountering the bump first. The rear wheel encounters the bump after a delay of wheelbase/velocity. Sprung mass displacement reflects combined front and rear effects. Pitch angle is positive for anticlockwise backward lean, when the front hits the bump and negative forward lean for the rear.



2.3.2 Roll analysis

Focusing on the front axle this 4-DOF model analyzes roll during cornering or uneven bumps. The left wheel hits the bump first, peaking early compared to right. Sprung mass shows heave motion with downward shift from left bump and oscillatory settling. Roll angle rises positively for initial left tilt, then negatively for right bump opposition, followed by damped oscillations.



2.4 Bicycle Model

The bicycle model simplifies vehicle longitudinal dynamics as a single-track system for straight-line motion, acceleration, and resistances. It assumes rear-wheel-drive, incorporating normal force distribution, aerodynamic drag/lift, tractive forces, and wheel rotation under pure rolling conditions.

2.5 Longitudinal Model

This extends the bicycle model by incorporating slip conditions for more accurate tractive forces. Front and rear tractive forces are derived from the magic formula using slip ratio inputs, forming a feedback loop. Unlike the

pure rolling assumption in the bicycle model, this accounts for slip to provide realistic values.

2.6 Full Car Actuation Model

The full car model integrates body motions (bounce, pitch, roll) and vertical movements of all four wheels for comprehensive performance analysis under road profiles. Equations assume clockwise positives for pitch and roll. A front anti-roll bar accounts for forces from wheel displacements and resistive torque during roll.

$$F_{fl} = k_f \left(x_{fl} - z - a\theta - \frac{t_{rf}\phi}{2} \right) + c_f \left(\dot{x}_{fl} - \dot{z} - a\dot{\theta} - \frac{t_{rf}\dot{\phi}}{2} \right)$$

$$F_{fr} = k_f \left(x_{fr} - z - a\theta + \frac{t_{rf}\phi}{2} \right) + c_f \left(\dot{x}_{fr} - \dot{z} - a\dot{\theta} + \frac{t_{rf}\dot{\phi}}{2} \right)$$

$$F_{rl} = k_r \left(x_{rl} - z + b\theta - \frac{t_{rr}\phi}{2} \right) + c_r \left(\dot{x}_{rl} - \dot{z} + b\dot{\theta} - \frac{t_{rr}\dot{\phi}}{2} \right)$$

$$F_{rr} = k_r \left(x_{rr} - z + b\theta + \frac{t_{rr}\phi}{2} \right) + c_r \left(\dot{x}_{rr} - \dot{z} + b\dot{\theta} + \frac{t_{rr}\dot{\phi}}{2} \right)$$

$$\delta_{fl} = x_{fl} - \left(z + a\theta + \frac{t_{rf}\phi}{2} \right)$$

$$\delta_{fr} = x_{fr} - \left(z + a\theta - \frac{t_{rf}\phi}{2} \right)$$

$$F_{arb,fl} = k_{arb,f}(\delta_{fl} - \delta_{fr})$$

$$F_{arb,fr} = -F_{arb,fl}$$

$$\delta_{rl} = x_{rl} - \left(z - b\theta + \frac{t_{rr}\phi}{2} \right)$$

$$\delta_{rr} = x_{rr} - \left(z - b\theta - \frac{t_{rr}\phi}{2} \right)$$

$$F_{arb,rl} = k_{arb,r}(\delta_{rl} - \delta_{rr})$$

$$F_{arb,rr} = -F_{arb,rl}$$

$$m_s \ddot{z} = F_{fl} + F_{fr} + F_{rl} + F_{rr} + F_{arb,fl} + F_{arb,fr} + F_{arb,rl} + F_{arb,rr} - m_s g - F_d$$

$$I_\theta \ddot{\theta} = a(F_{fl} + F_{fr} + F_{arb,fl} + F_{arb,fr}) - b(F_{rl} + F_{rr} + F_{arb,rl} + F_{arb,rr})$$

$$I_\phi \ddot{\phi} = \frac{t_{rf}}{2}(F_{fl} + F_{arb,fl} - F_{fr} - F_{arb,fr}) + \frac{t_{rr}}{2}(F_{rl} + F_{arb,rl} - F_{rr} - F_{arb,rr})$$

$$m_u \ddot{x}_{fl} = k_t(z_{fl} - x_{fl}) + c_t(\dot{z}_{fl} - \dot{x}_{fl}) - F_{fl} - F_{arb,fl} - m_u g$$

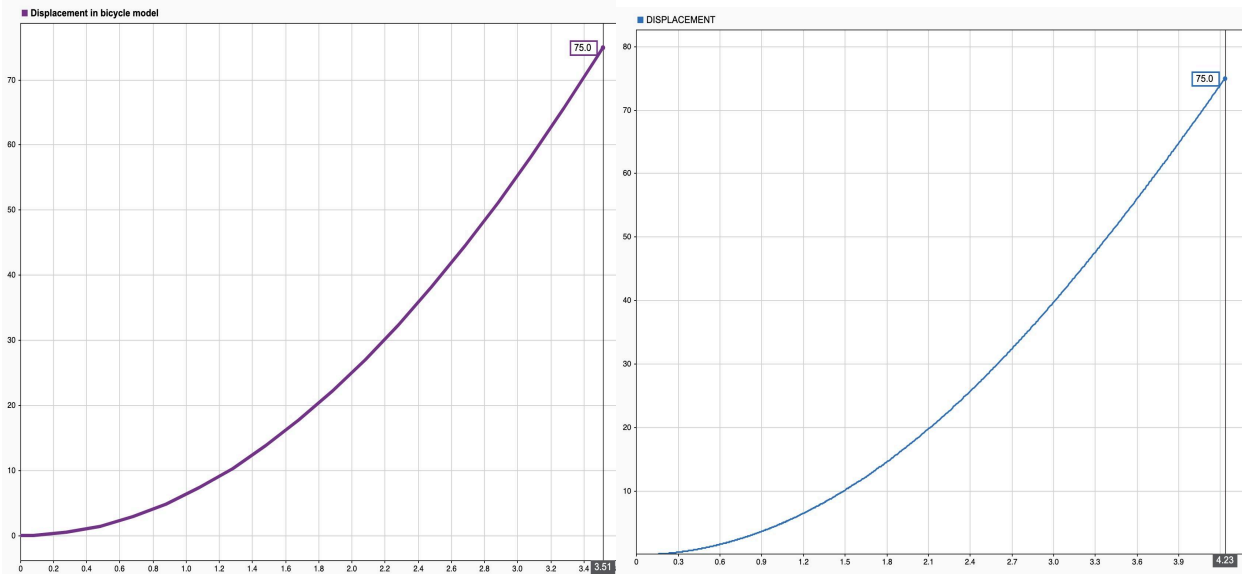
$$m_u \ddot{x}_{fr} = k_t(z_{fr} - x_{fr}) + c_t(\dot{z}_{fr} - \dot{x}_{fr}) - F_{fr} - F_{arb,fr} - m_u g$$

$$m_u \ddot{x}_{rl} = k_t(z_{rl} - x_{rl}) + c_t(\dot{z}_{rl} - \dot{x}_{rl}) - F_{rl} - F_{arb,rl} - m_u g$$

$$m_u \ddot{x}_{rr} = k_t(z_{rr} - x_{rr}) + c_t(\dot{z}_{rr} - \dot{x}_{rr}) - F_{rr} - F_{arb,rr} - m_u g$$

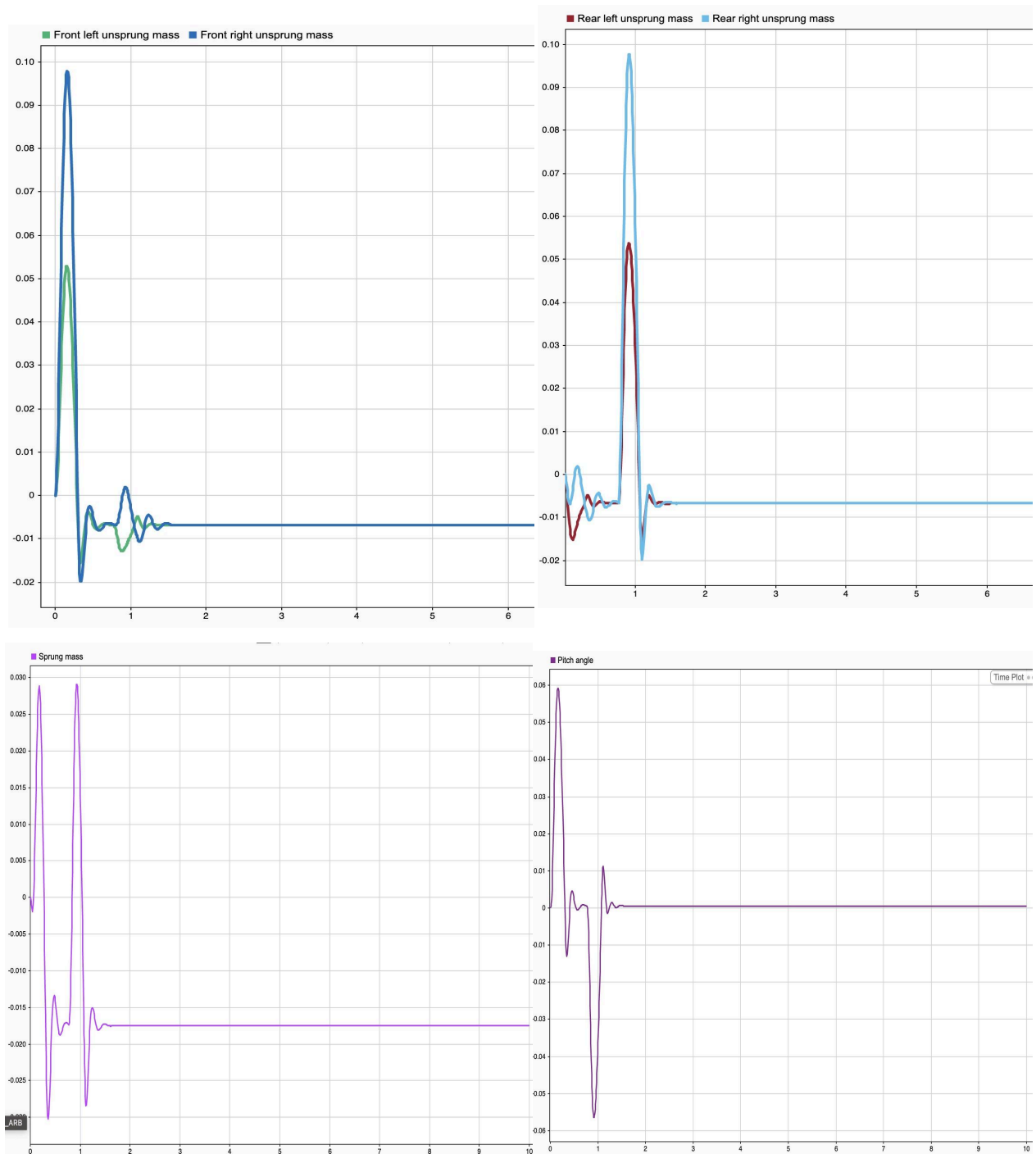
SIMULATION RESULT

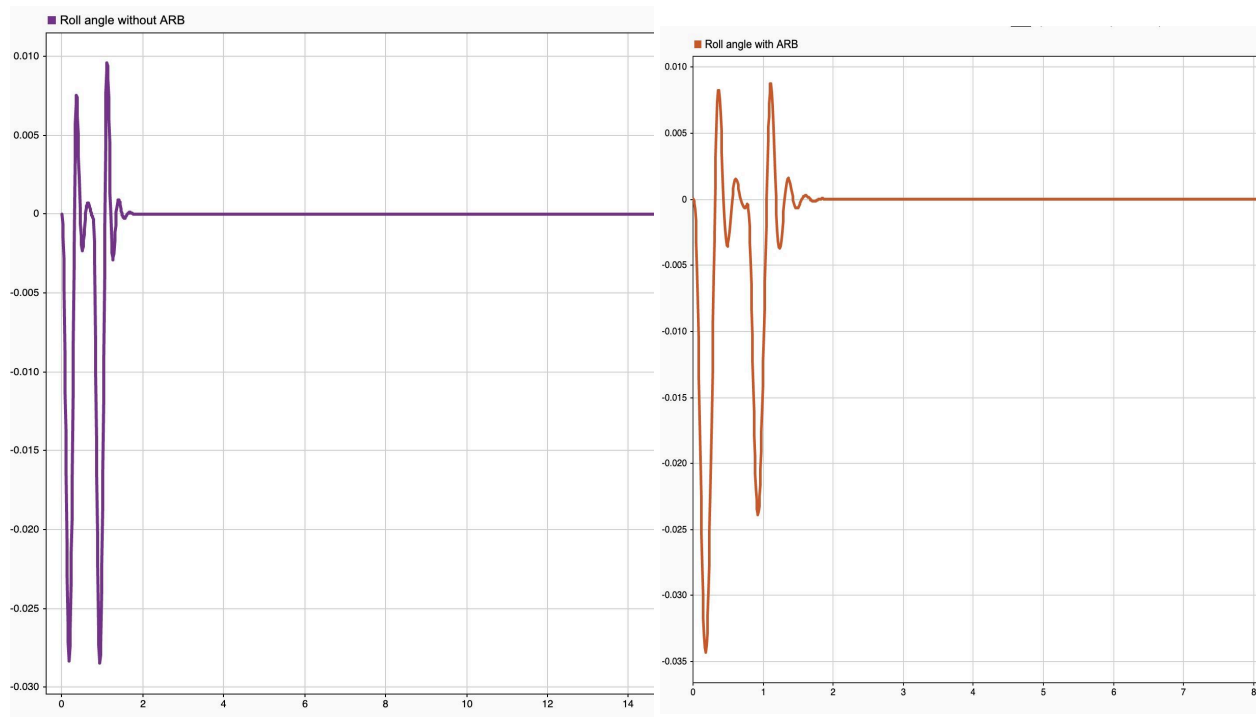
Bicycle and longitudinal model have been used to find the time taken to cover 75m. The time measured through bicycle model is 3.51s and by the longitudinal model is 4.23s. The longitudinal model gave more accurate result compared to the bicycle model since the condition of pure rolling is not assumed on longitudinal model.



With the Road input of Sine wave of width 0.3 m, left front and rear height 0.05 m, right 0.1 m for roll induction and rear delay by wheelbase/velocity.

Right wheels show higher displacement due to taller bumps. Sprung mass peaks positively when crossing bumps, with negative impacts post-crossing. Pitch shows positive with backward lean for front bump and negative for rear. Roll is negative that is anticlockwise due to higher right bump, with oscillatory peaks.





CONCLUSION

In this project, a comprehensive 7-DOF full car actuation model for an FSAE vehicle was developed, incorporating sprung/unsprung masses, suspension components, a front anti-roll bar, and aerodynamic influences. Simulations revealed how the anti-roll bar mitigates roll amplitude, while camber and toe variations significantly impact tire contact, grip, wear, and overall performance. The model predicts tire wear in endurance tests and, when tuned, provides lateral force for skid pad analysis to determine cornering speeds. Progressive modeling from quarter to full car enabled detailed analysis of bounce, pitch, roll, and wheel movements under dynamic conditions.

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[Simulation and Analysis of Full Car Model for various Road profile on a analytically validated MATLAB/SIMULINK model](#)

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[Full Vehicle Model - an overview | ScienceDirect Topics](#)

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